Q1. Define research and explain its characteristics

Definition of Research

- Research is a systematic and scientific process of investigating a problem to discover new knowledge, verify existing facts, or develop new theories.
- Clifford Woody: *"Research comprises defining and redefining problems, formulating hypothesis, collecting, organizing and evaluating data, making deductions and reaching conclusions."*
- Creswell: *"Research is a systematic investigation to establish facts."*

👉 In simple words, **research = re (again) + search (to find)** → it means a careful study to gain new understanding or verify existing knowledge.

Characteristics of Research

1. **Generalized** – Findings should be applicable to a larger population, not just the sample studied.
2. **Controlled** – Research tries to control influencing variables to establish cause–effect relationships.
3. **Rigorous** – Procedures are relevant, appropriate, and carefully followed.
4. **Empirical** – Based on observable evidence and real-life data.
5. **Systematic** – Conducted in a logical sequence, step by step.
6. **Reliability** – Results should be consistent if repeated under similar conditions.
7. **Validity & Verifiability** – Findings should measure what they claim and be verifiable by others.
8. **Hypothesis-based** – Begins with assumptions/hypotheses to test.
9. **Analytical & Accurate** – Data should be critically examined and error-free.
10. **Credible** – Research must use authentic sources and proper references.
11. **Critical** – Should withstand scrutiny and evaluation.

✍️ **Exam writing tip:** Start with definition → explain meaning → list at least 6–7 key characteristics with short examples.

Q2. What are the objectives of research?

The objectives of research show **why research is undertaken**. According to your notes, the main objectives are:

1. **To gain knowledge** – Discover new facts and develop new insights.
2. **To understand and explain phenomena** – Describe and explain "what, why, and how" things happen.

3. **To verify existing knowledge** – Test theories, concepts, and assumptions.
4. **To solve problems** – Provide practical solutions to social, economic, or scientific issues.
5. **To predict outcomes** – Anticipate future trends/events using reliable data.
6. **To aid in decision-making** – Help policymakers, managers, and organizations take better decisions.
7. **To develop new theories and models** – Contribute to academic and scientific development.
8. **To improve practices** – Find better ways of doing tasks (e.g., improving teaching methods, business strategies, or healthcare practices).

👉 In exams, write: "The general objectives of research are knowledge generation, problem-solving, verification of existing facts, prediction, and theory development."

Q3. Discuss different types of research with examples

Research can be classified into different categories:

1. Based on Application

- **Basic/Pure/Fundamental Research** → expands existing knowledge without immediate practical use.
  Example: Developing a new theory in physics.
- **Applied Research** → solves specific practical problems.
  Example: Research to improve a hotel's customer service.

2. Based on Objectives

- **Descriptive Research** → describes a situation or phenomenon.
  Example: Survey of customer preferences in restaurants.
- **Correlational Research** → identifies relationships between variables.
  Example: Studying link between study hours and exam performance.
- **Explanatory Research** → explains why events occur.
  Example: Research on why customers shift from local shops to malls.
- **Exploratory Research** → initial study when problem is not well-defined.
  Example: A startup exploring if demand exists for a new food product.

3. Based on Inquiry Mode

- **Quantitative Research** → focuses on numbers, measurable data.
  Example: Measuring hotel occupancy rates.
- **Qualitative Research** → focuses on meanings, experiences, and opinions.
  Example: Interviews about guest satisfaction.
- **Mixed Method** → combination of both.

4. Other Classifications

- **Descriptive vs. Analytical**
- **Conceptual vs. Empirical**
- **Experimental vs. Non-experimental**

Q4. Differentiate between Conceptual and Theoretical Models

Conceptual Model

- A **conceptual model** is built around **concepts, assumptions, or ideas** that explain a phenomenon.
- It is **based on understanding, imagination, and logical reasoning**, not necessarily tested scientifically.
- Used mainly by **thinkers and philosophers** to develop new concepts or refine existing ones.
- **Example:** Maslow's hierarchy of needs (conceptual framework to explain human motivation).

Theoretical Model

- A **theoretical model** is built upon **existing theories, established principles, and research evidence**.
- It is **scientific, tested, and validated**; aims to explain relationships between variables.
- Helps in formulating hypotheses and guiding research methodology.
- **Example:** Herzberg's Two-Factor Theory (theory-based model explaining job satisfaction).

Key Differences (Tabular Form)

Basis	Conceptual Model	Theoretical Model
Nature	Abstract, idea-based	Evidence-based, derived from theories
Source	Philosophers, thinkers	Established research, tested knowledge
Validation	Not always tested scientifically	Supported by data and experiments
Purpose	To develop new concepts or refine ideas	To test, explain, and predict phenomena
Example	Maslow's Need Hierarchy	Herzberg's Motivation Theory

👉 **Exam tip:** Always write definitions + 3–4 clear points of difference (preferably in a table) + one example.

Q5. Discuss the importance of research methodology in scientific research

Meaning

- **Research methodology** = the **systematic process, methods, and techniques** used to conduct research.

- It answers *"how the research is done"* → what methods, tools, and steps are followed.

Importance in Scientific Research

1. **Provides a systematic approach** – Ensures research is carried out logically and step by step.
2. **Ensures reliability & validity** – Proper methodology gives consistent and accurate results.
3. **Helps in problem formulation** – Guides the researcher in defining the problem clearly.
4. **Selects correct methods** – Helps in choosing appropriate tools (qualitative/quantitative).
5. **Improves accuracy** – Minimizes bias and errors in data collection & analysis.
6. **Facilitates comparison & replication** – Other researchers can repeat the study using the same methodology.
7. **Enhances credibility** – Well-designed methodology increases trustworthiness of findings.
8. **Decision-making tool** – Provides dependable data for managers, scientists, and policymakers.
9. **Supports theory building** – Helps validate or develop new theories in science.

👉 Without research methodology, scientific research would lack **clarity, consistency, and trustworthiness**.